

How-to · Chrome Prompt API

A Wikipedia chatbot that runs entirely in your browser

A full RAG pipeline in ~300 lines of TypeScript. No backend.

Context

The browser quietly became an inference target

- Chrome ships `window.LanguageModel`
- An on-device LLM, callable from a `<script>` tag
- No API key. No fetch. No bill.
- Inference moves to the client.

Why it matters

On-device inference changes the economics

- No per-token cost
- No telemetry by default
- No *"we're sorry, the model is overloaded"* page
- The browser is now a deploy target for LLMs

Problem

Tiny local models forget facts

Great at language. Terrible at trivia.

They produce plausible nonsense with extreme confidence — wrong dates, wrong people, wrong acquisitions.

Insight · RAG

Make the model read, not remember

- Fetch authoritative text at query time
- Stuff it into the prompt
- The model becomes a reading-comprehension engine
- A job it's actually good at

Insight · Corpus

Why Wikipedia is a near-perfect demo corpus

- Huge, broad, human-curated, current
- Clean public REST API
- No keys, no quotas, no scraping
- No licensing acrobatics for a zero-backend project

Insight · Pipeline

A full RAG loop in 5 steps

1. User asks a question
2. Disposable session distills it into a search query
3. Wikipedia returns the top 3 article keys
4. Each summary is fetched in parallel
5. Persistent session streams a grounded answer

Insight · The trick

Two sessions, two jobs

- **Disposable** — rewrites the query, no history, destroyed after one use
- **Persistent** — carries the conversation, resolves *"where did she work?"*
- Same model. Very different roles.

Tradeoff · Context

Use the summary, not the full article

On-device context is tight. A 50k-token article doesn't help — it makes the model slower and loses the user's question.

Wikipedia editors already wrote the lead paragraph for you.

Insight · Security

Model output is untrusted input

- Stream → `marked` → `DOMPurify` → DOM
- A model can emit `<img src=x onerror=...>` by accident
- Two lines of code stop a Markdown chat from getting XSSed by itself

Apply

The shape generalizes

- Swap Wikipedia for product docs → in-browser docs assistant
- Swap for a public API → distributable domain chatbot, static site
- Swap for IndexedDB notes → private journal that never phones home

Next step

Clone it. Point Chrome at it. Ask it something it doesn't know.

github.com/jeromeetienne/prompt_api_wikipedia